

Discovering and defining subdomains

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Founder of Eventuate.io

Founder of the original CloudFoundry.com

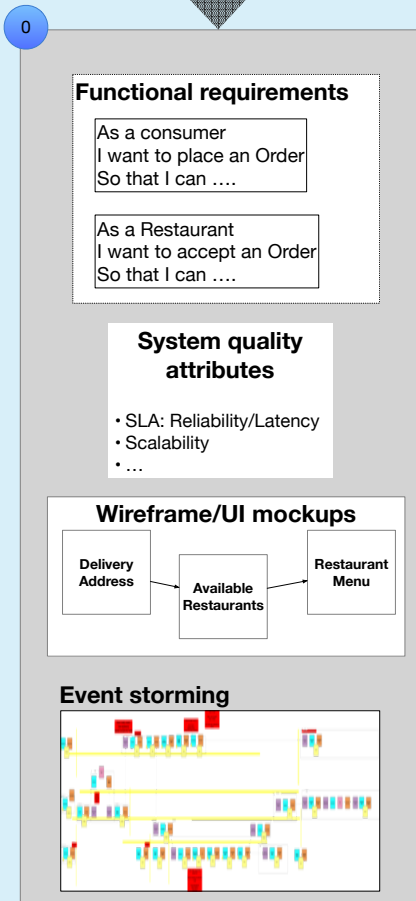
Author of POJOs in Action and Microservices Patterns

 @crichardson

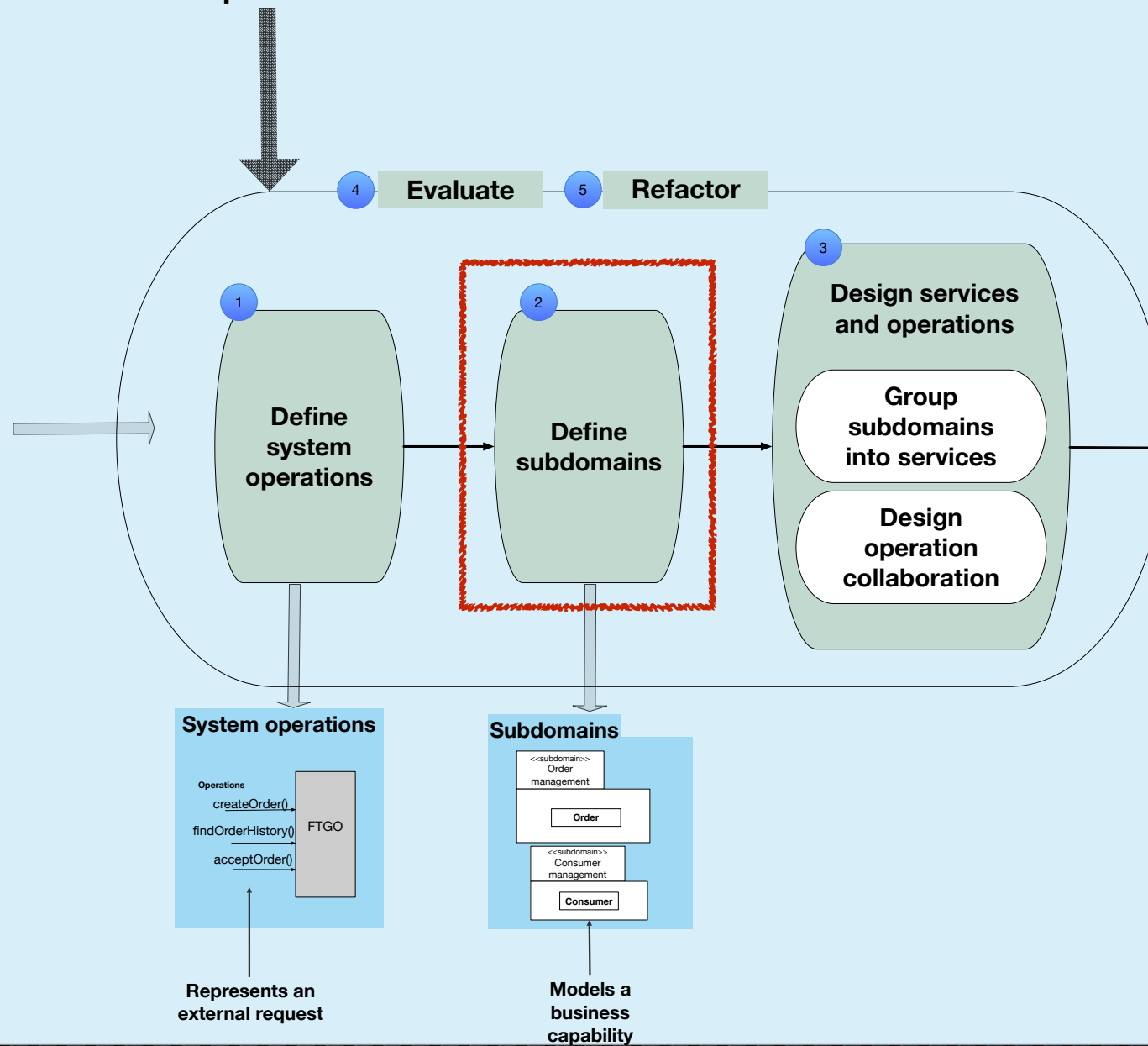
chris@chrisrichardson.net

adopt.microservices.io

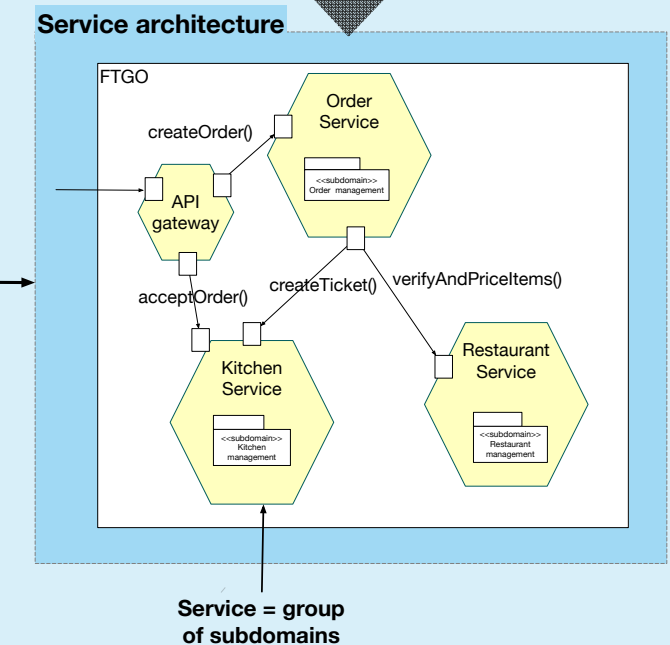
The starting point



The process



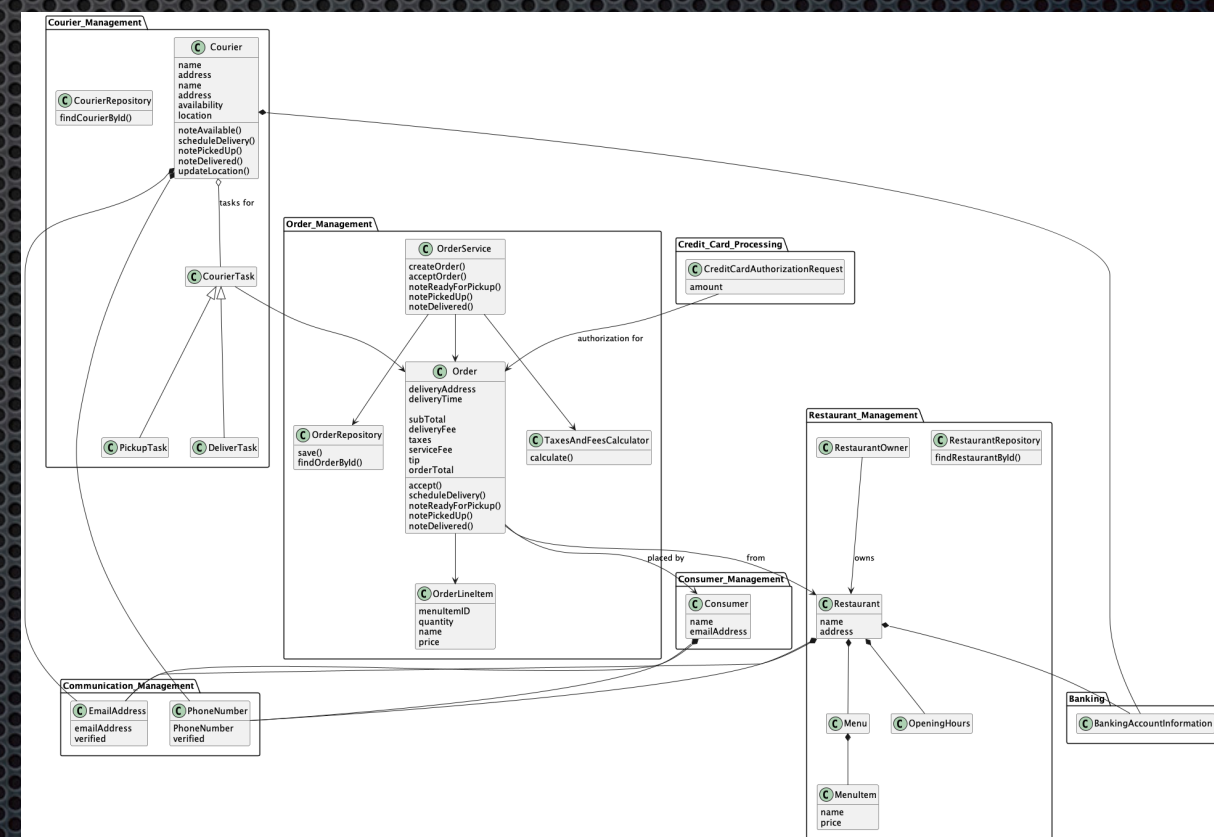
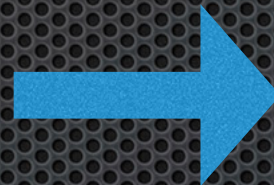
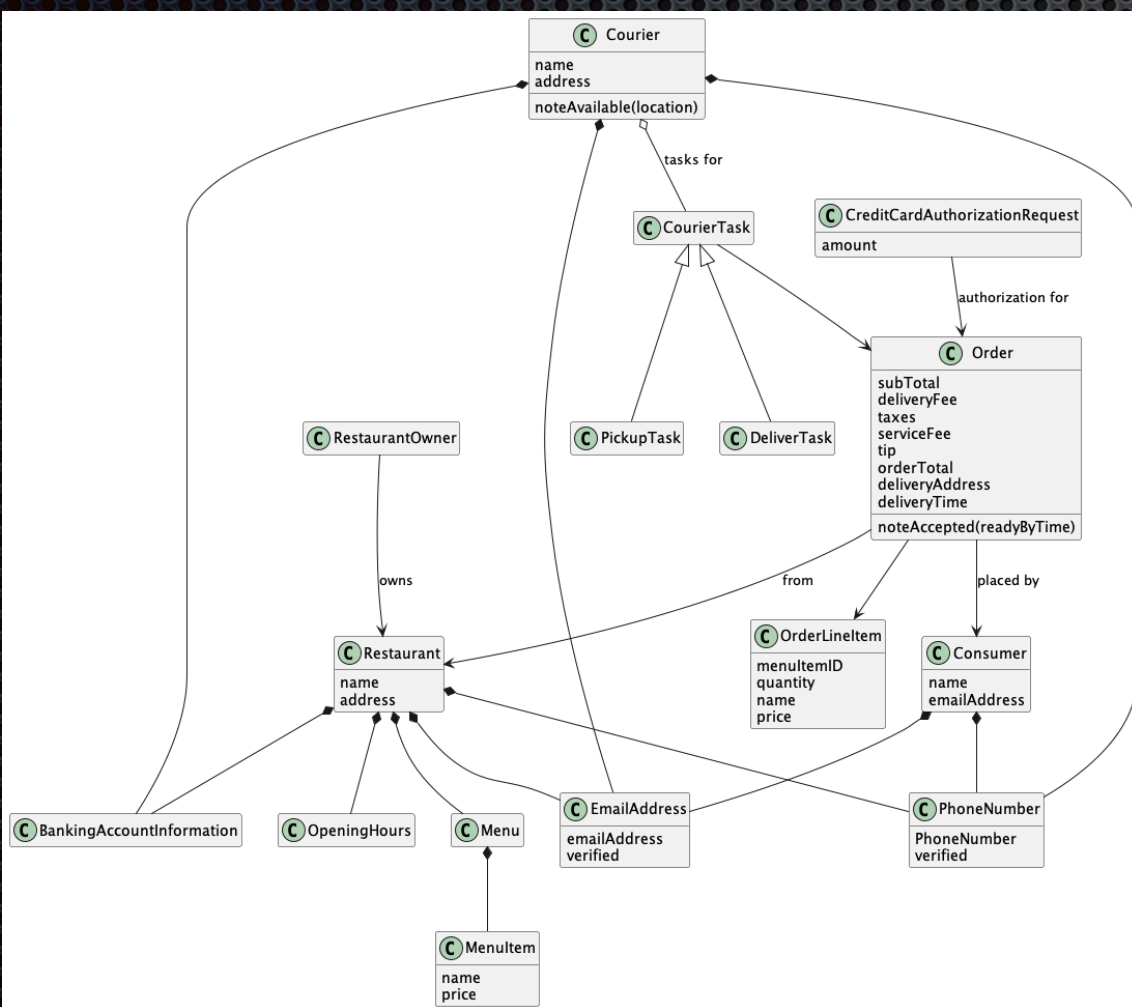
The result



Agenda

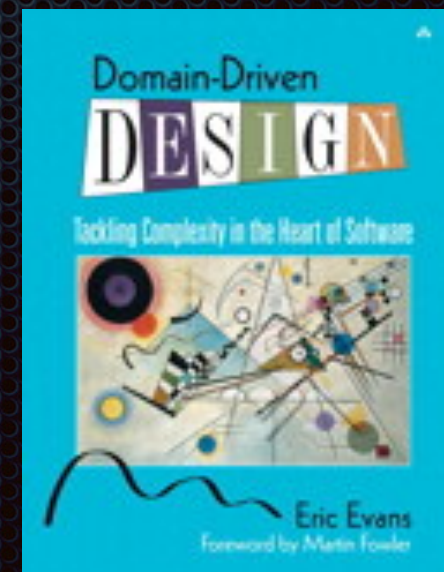
- Overview of subdomains
- Discovering subdomains

Refine business entities and organize into subdomains



About subdomains

- Domain is what your business does, e.g. online retailer
- Subdomain is a functional area of the business
- Often corresponds to the organization chart
- For example:
 - Order management
 - Restaurant management
 - Delivery management
 - ...
- Subdomains are grouped to form services



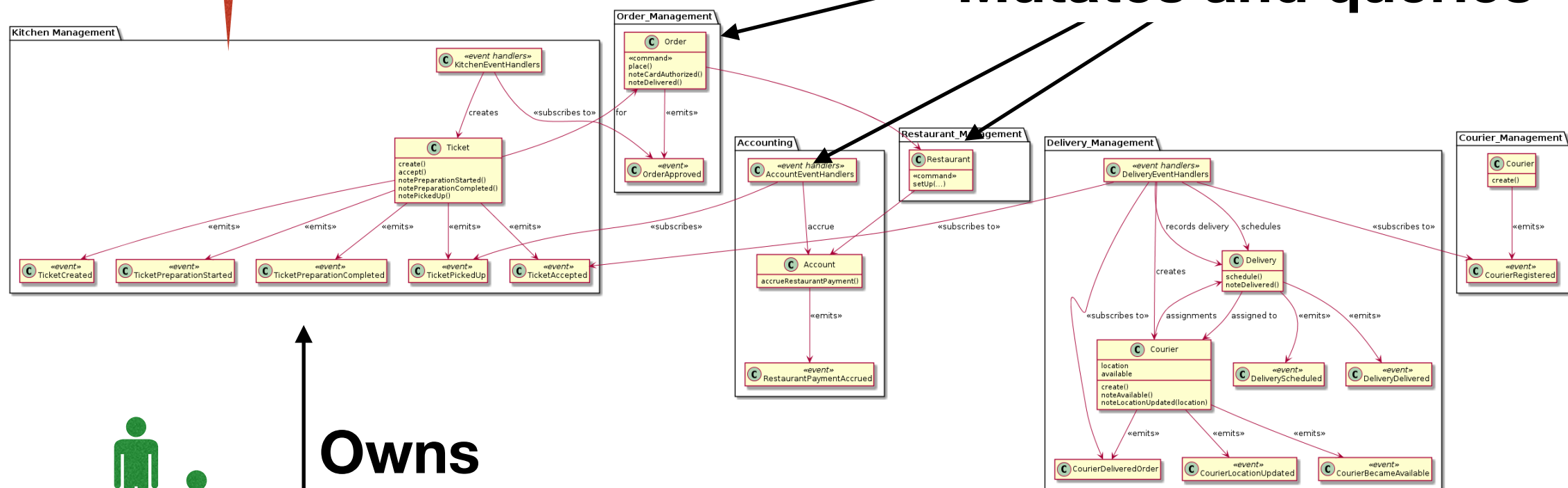
Subdomains vs. system operations

Models/implements business capability

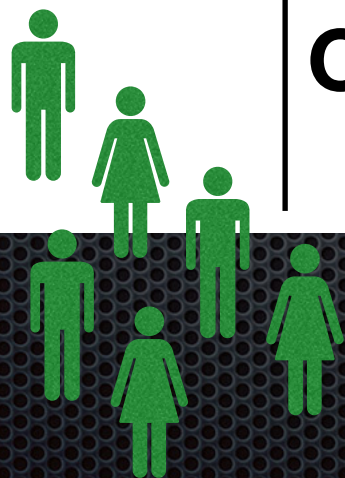
createOrder()

FTGO Application

Mutates and queries



Owns



Kitchen Team

High-level view:

subdomain = collection of aggregates

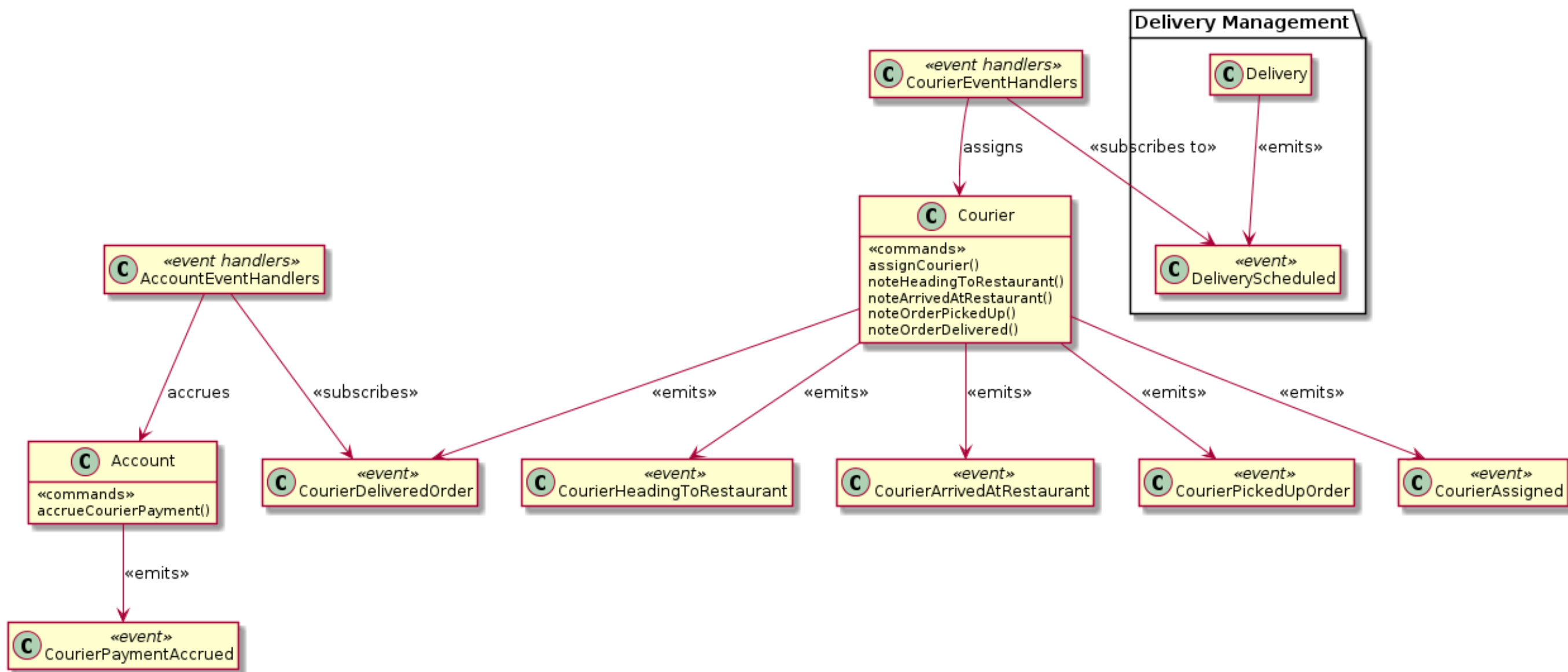
- Customer Management: Customer
- Restaurant Management: Restaurant
- Order Management: Order
- Delivery Management: Delivery, Courier
- ...

Documenting subdomain: canvas...

Subdomain canvas	
Name	Delivery Management
Responsibilities	• Scheduling deliveries
Collaborations	• Scheduling deliveries • Knowing Courier's location and availability • Managing Courier's pickup and delivery schedule • ...
Type	DDD: core, supporting, generic
Complexity	Cynefin: simple, complicated, complex
Business criticality	Tier 1, tier 2, ...
Regulations	e.g. PCI, GDPR, SamD

Dark energy:
Segregate by
characteristics

... Documenting subdomains: aggregates:



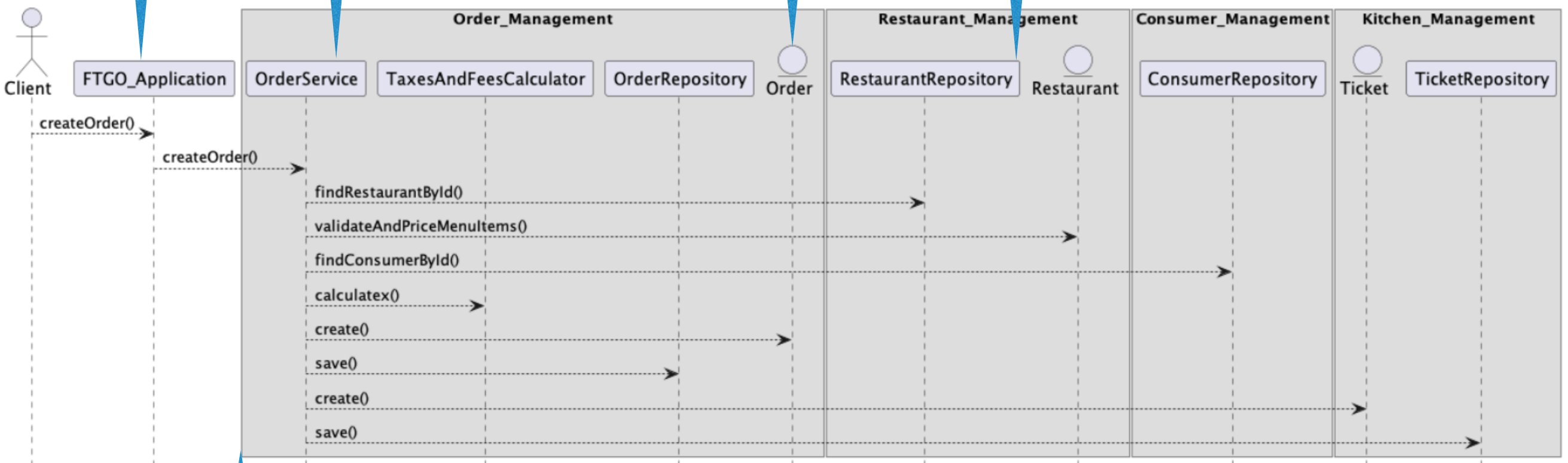
... Documenting subdomains: collaboration diagrams

Optional

Optional

Very useful

Optional



Subdomain

Note:

- OrderService is a class
- Order_Management is a subdomain
- ...

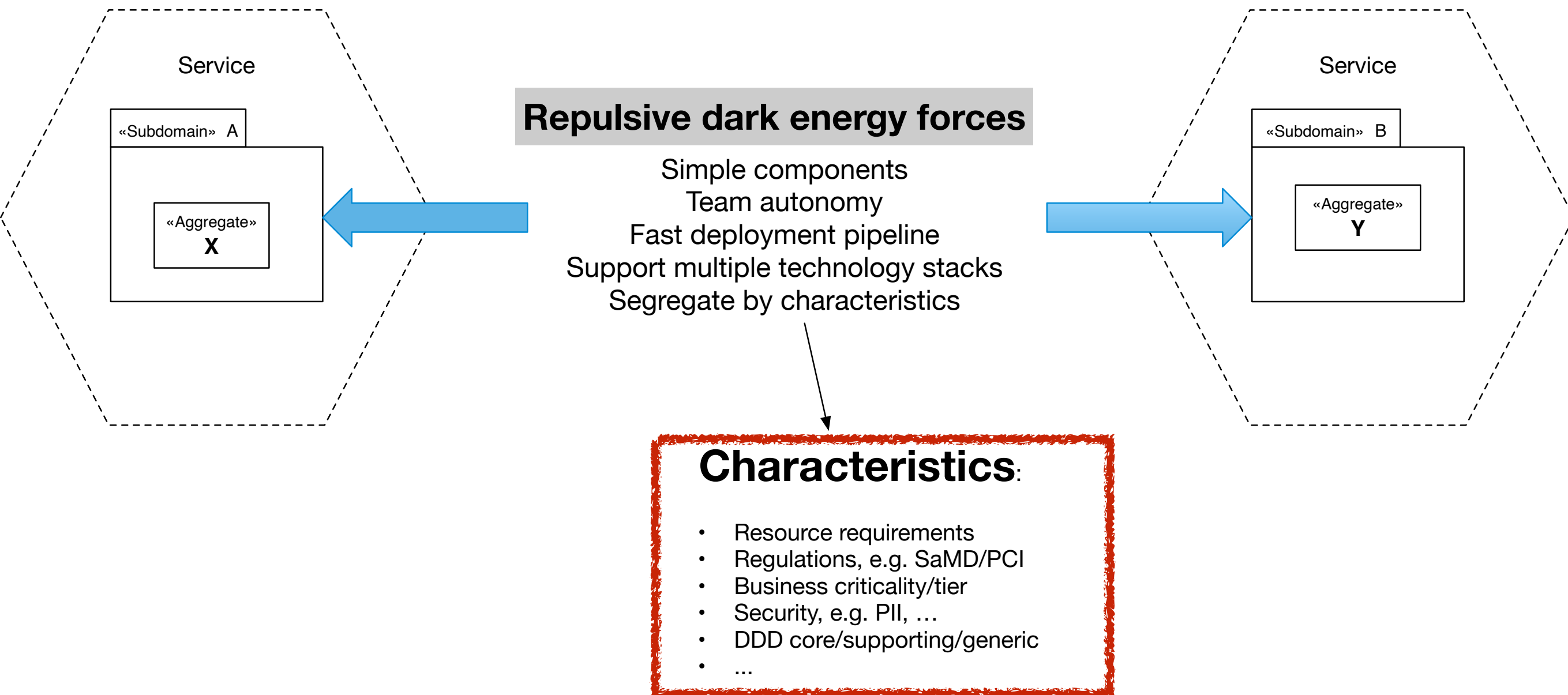
Use DDD concepts

But

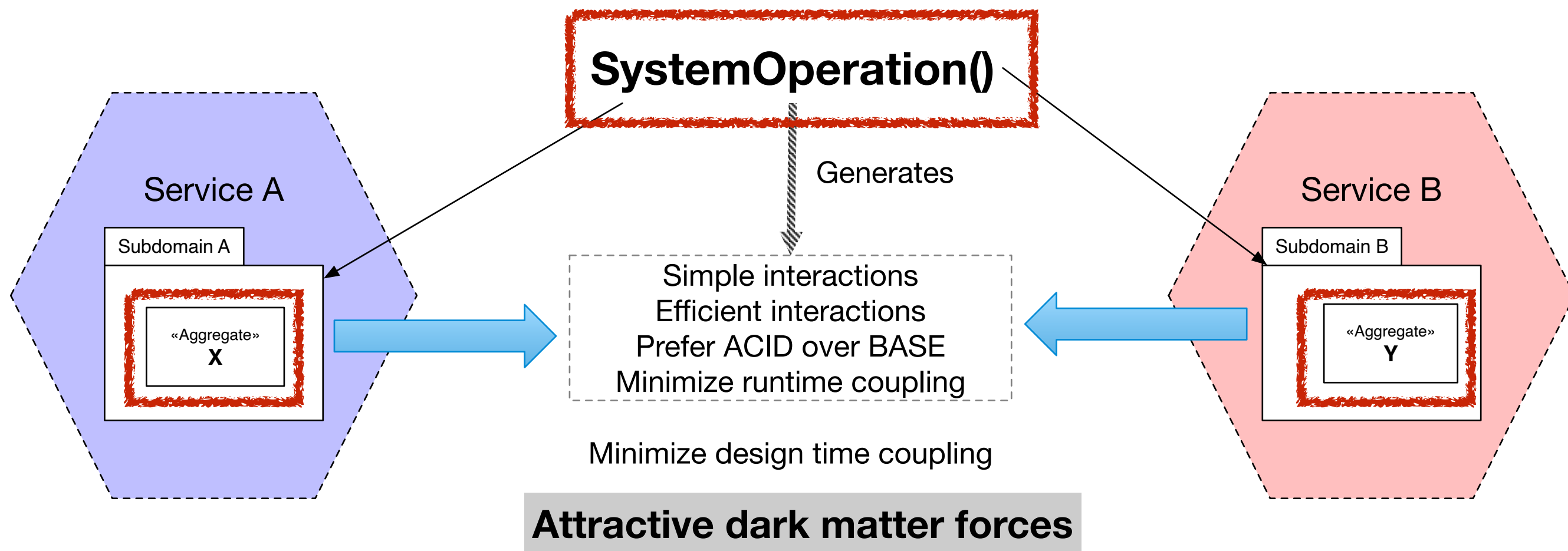
Don't need detailed models

Just enough modeling in order
to define the service architecture

Focus: subdomain characteristics



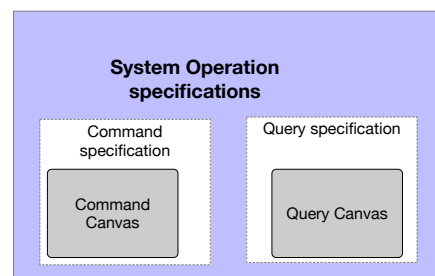
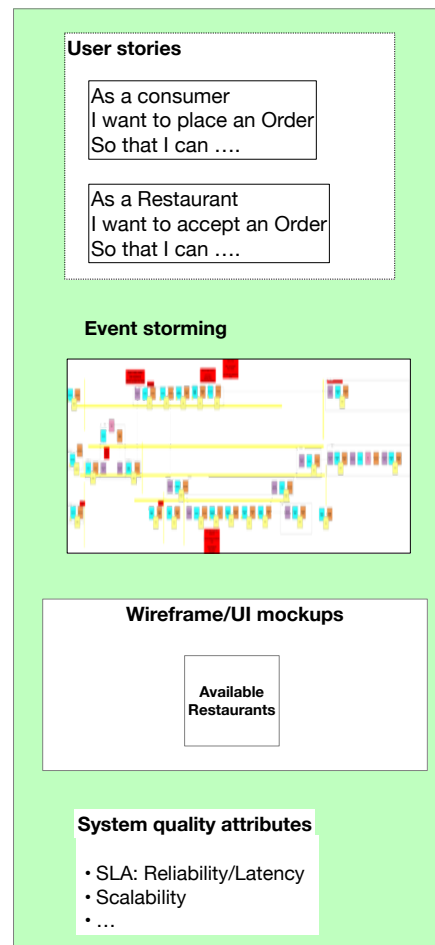
Focus: subdomain aggregates



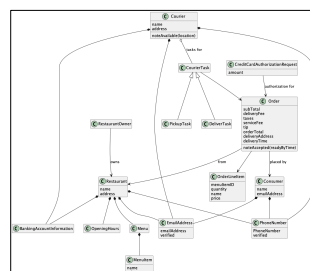
Agenda

- Overview of subdomains
- Discovering subdomains

How to define subdomains



Initial model



Define subdomains

Discover candidate subdomains

Analyze requirements, domain language, etc.

Analyze Event storming Swim Lanes and Pivotal events

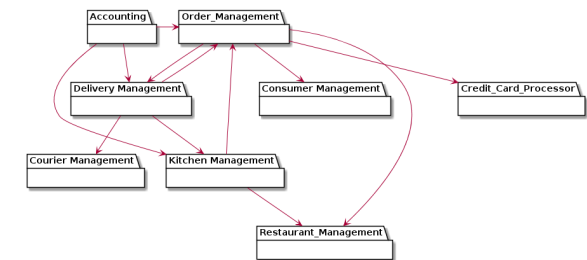
Analyze Business Architecture

Organize classes into subdomains

Refine

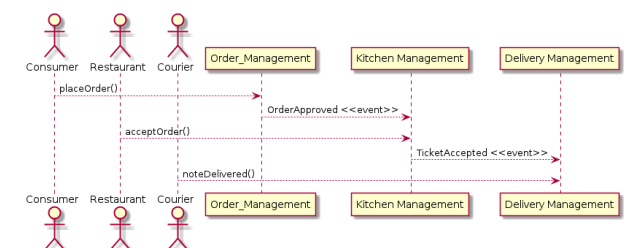
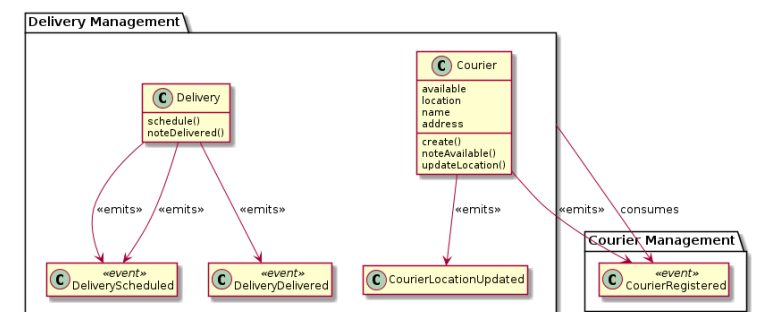
Refactor and Distill

Analyze, And Transform



Subdomain canvas

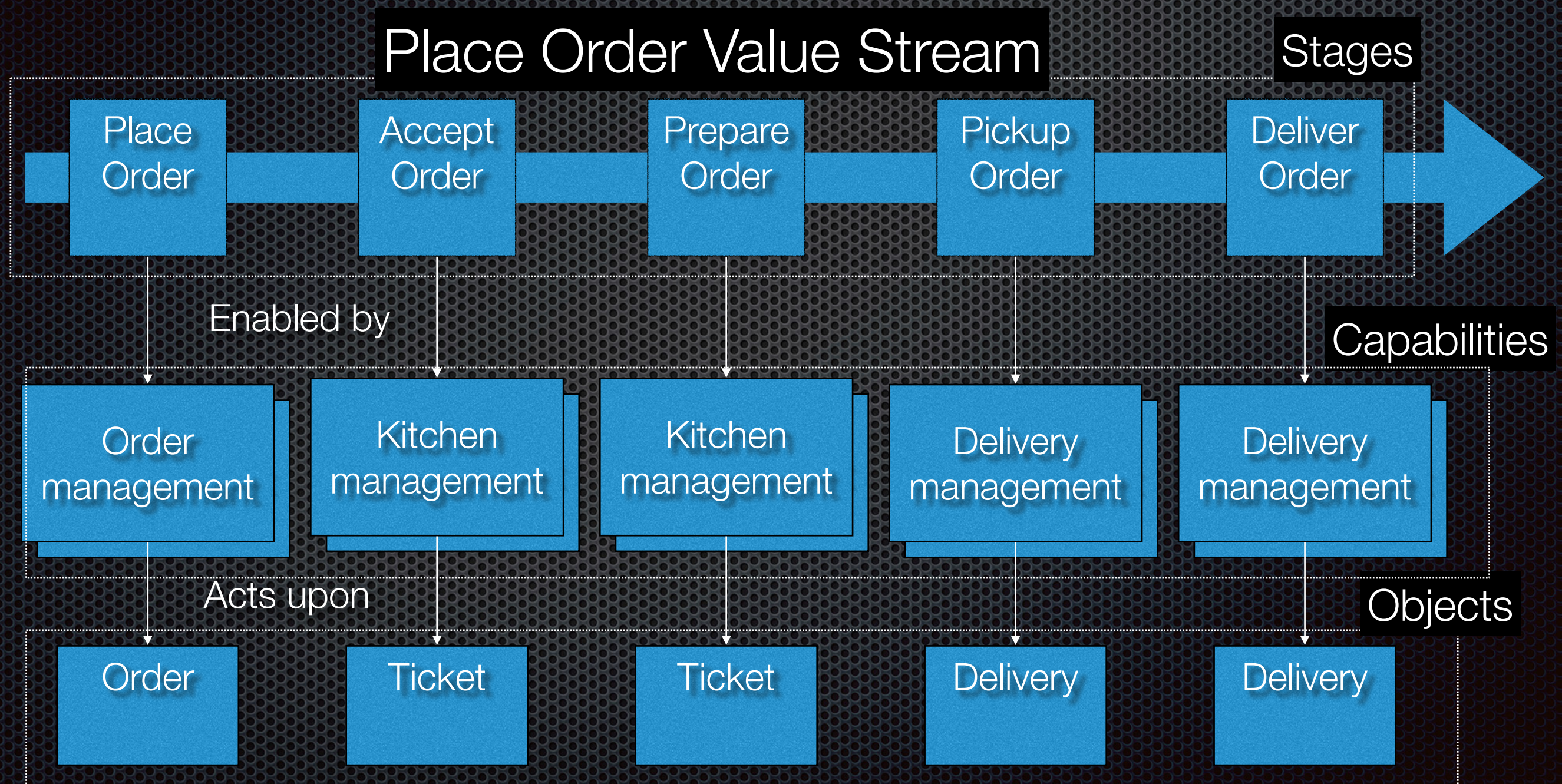
Name	Restaurant Management
Responsibilities	• Manages information about restaurants • Knows restaurant name, location, description, opening hours, and menu
Collaborations	• External address validation and geocoding service
Type	Supporting
Complexity	Simple
Key aggregates	• Restaurant



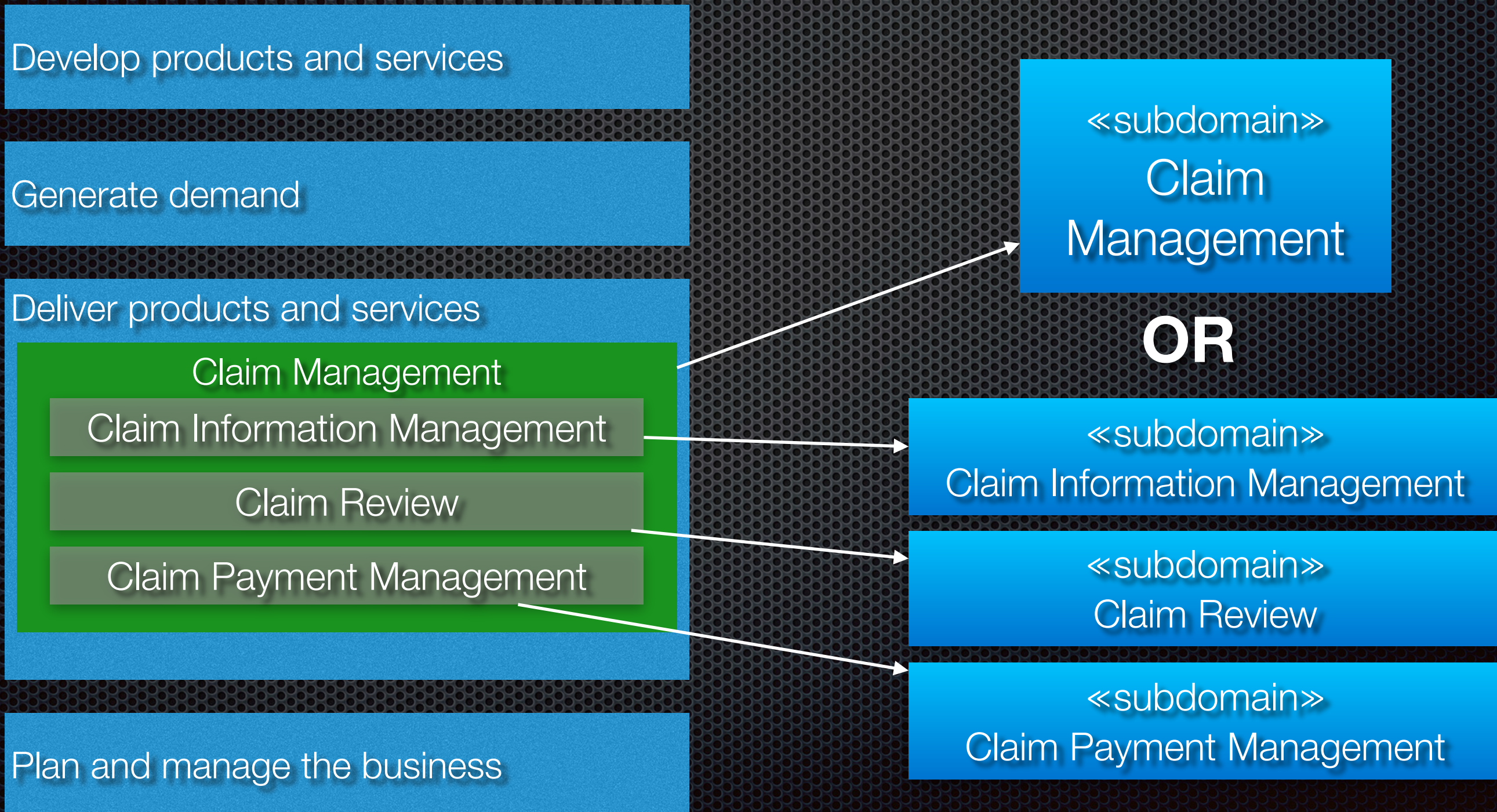
Analyze the domain (business)

- Extract nouns and verbs from requirements
- Talk to domain experts
- Build subdomains:
 - Map nouns to classes
 - Verbs to methods
- Identify subdomain boundaries
 - Linguistic boundaries: e.g. same noun has different meanings
 - Areas of expertise
 - Organization structure
- ...

Business architecture: value streams that deliver value



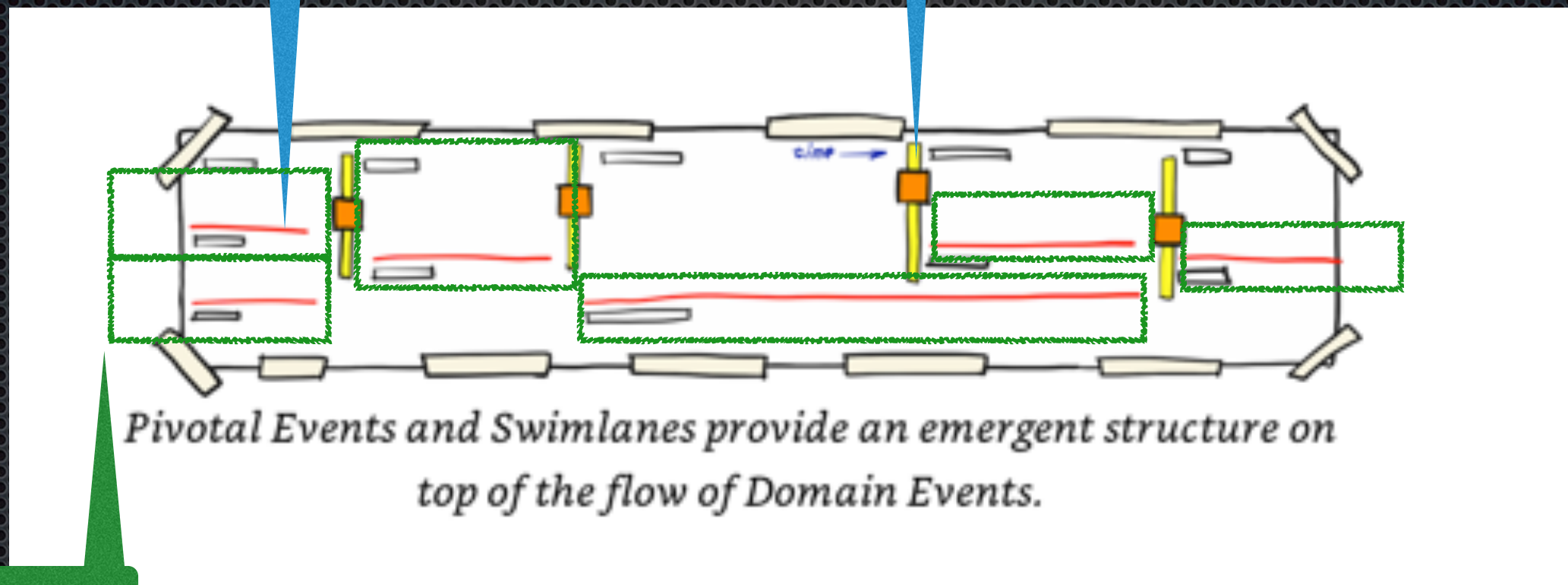
Map business capabilities to subdomains



Use event storming to identify subdomains

Swimlane = separate but interconnected flows

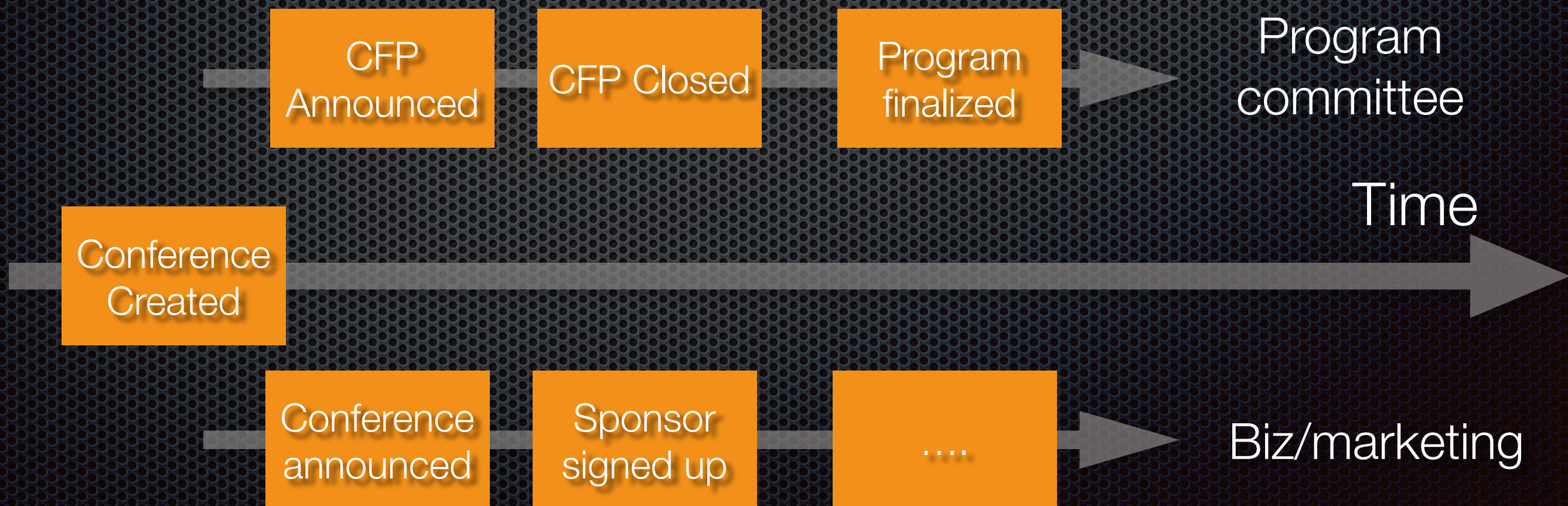
Pivotal event = transition between business phases



Subdomain

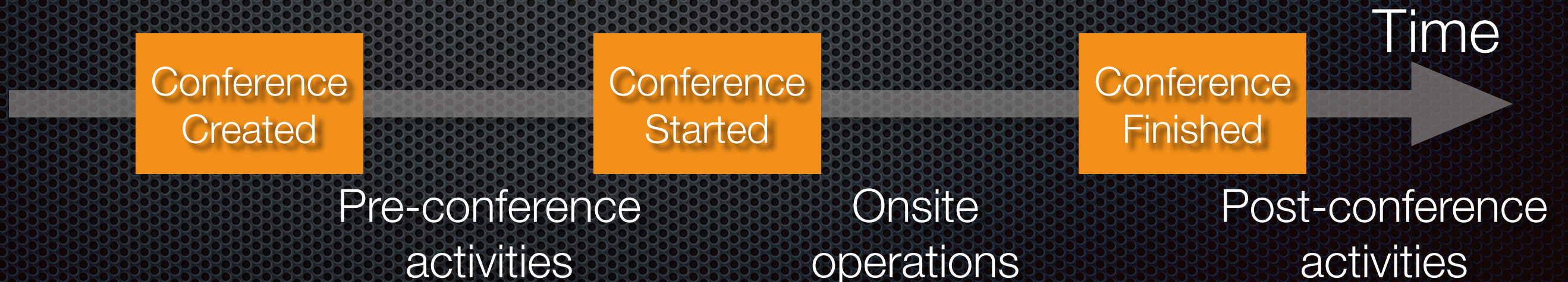
About swim lanes

- Swimlanes are loosely coupled, independent or parallel flows
- Often activities performed by different departments = different behavior = potentially different subdomains

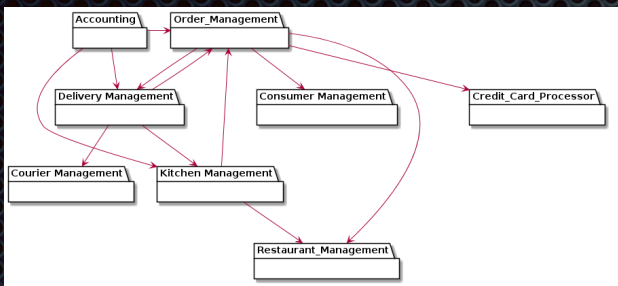


About pivotal events

- Pivotal event represents a major transition in the lifecycle of business entity
- For example: Conference Started, Conference Finished,
- The phase between two pivotal events = different behavior = potentially different subdomains



Summary: deliverables



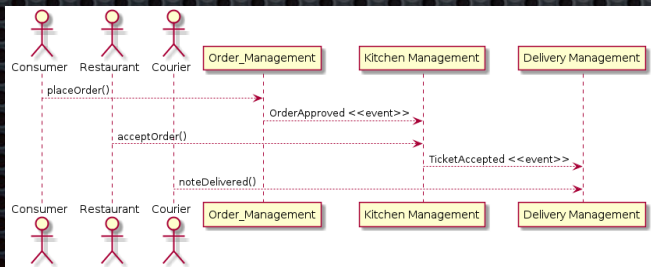
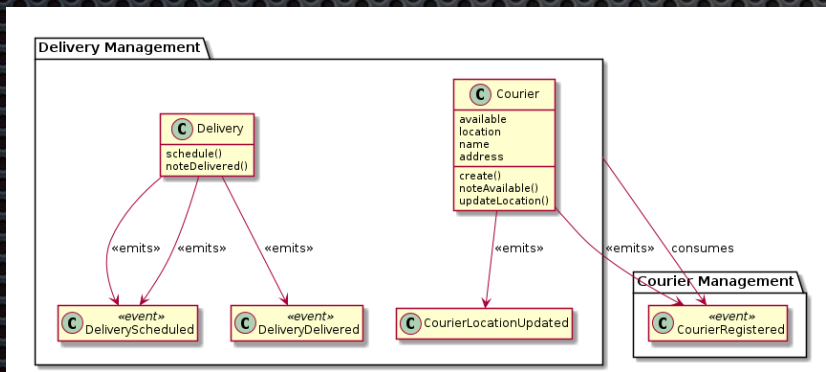
Revised

System command canvas

Name	placeOrder(consumerID, deliveryTime, deliveryAddress, restaurantID, lineItems)		
Parameters	• consumerID - the ID of the consumer placing the order • deliveryTime - the desired delivery time • deliveryAddress - the delivery address • restaurantID - the restaurant • lineItems - collection of OrderLineItem(menuItemIDs, quantity)		
Return value	• orderId		
Trigger	User command		
Service Objectives	<table><tr><td>Level</td><td> • Business tier: Critical, Core, User, Internal • Throughput, e.g. requests/second • Availability - ... • Latency - ... • ... </td></tr></table>	Level	• Business tier: Critical, Core, User, Internal • Throughput, e.g. requests/second • Availability - ... • Latency - ... • ...
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Preconditions	• a Consumer with ID consumerID exists • the Consumer has a valid payment method • a Restaurant with ID restaurantID exists • the deliveryAddress and deliveryTime are served by the restaurant • MenuItem's identified by menuItemID exists		
Postconditions	• an Order order was created in the APPROVED state • a CreditCardAuthorization for the orderTotal for the Consumer's credit card was created • returnValue became order.ID		

Subdomain canvas

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Type	Supporting
Complexity	Simple
Key aggregates	• Restaurant



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Questions?

adopt.microservices.io

About:

Kata #2: Define subdomains

Define and document the subdomains using the templates in 02-defining-subdomains

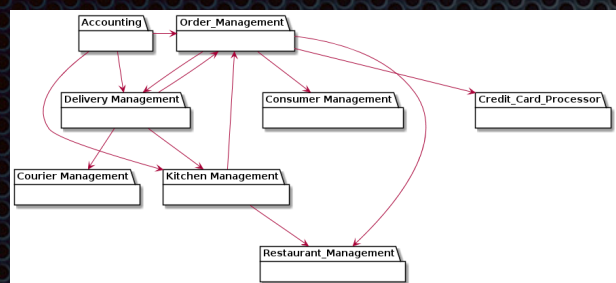
- As a group:
 - Discover and define subdomains
 - Assign system operations to pairs (or maintain a task queue)
- In pairs:
 - Design each system operation collaborations in terms of entities and subdomains
- As a group:
 - Review and *harmonize* system operation collaborations
 - Refine subdomains

Template: 02-defining-subdomains

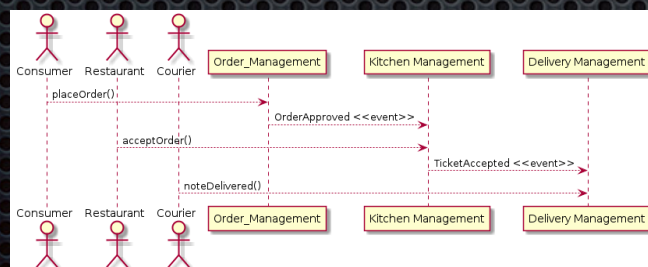
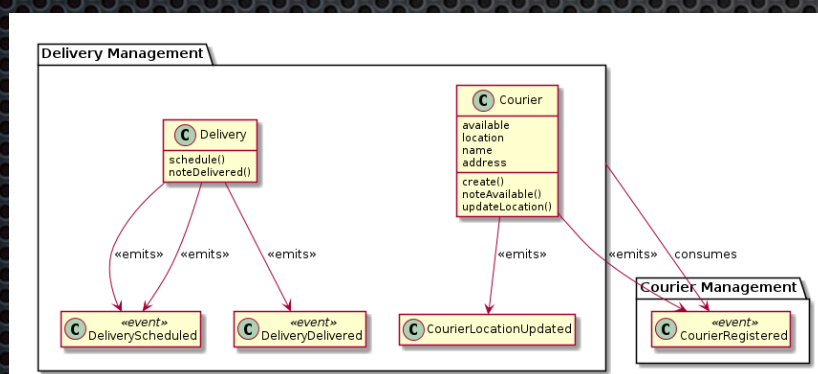
Key model behavior

- Keep track of the number of rooms of each room type that are available for sale on a given date
- Handle the following *make reservation* scenario:
 - Only one room of a room type available at a given price
 - Two simultaneous requests for the same room type
 - One should succeed, one should fail (get higher price)

Summary: deliverables



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Revised

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Template: 02-defining-subdomains json